

Homework Sheet — Stage 7: Outliers + Risk Assumptions

All paths below are inside `homework/homework07/`.

Assignment

Implement reusable functions to detect and handle outliers, run a simple sensitivity analysis, and reflect on assumptions.

Chain: In the lecture, we learned how to detect, remove, or adjust outliers (IQR, Z-score, winsorizing) and to test their impact on simple models. Now, you will adapt these methods to a provided dataset to assess how your choices affect results and to document risks and assumptions.

What You'll Do

1. Copy `stage07_outliers-risk-assumptions_homework-starter.ipynb` into `homework/homework07/`, rename it `homework07_outliers-risk-assumptions_submission.ipynb`, and open it.
2. Run the **Setup** cell at the top of the starter notebook. It generates the needed `.csv` for you and saves it under `data/raw/` - no dataset is handed out for this assignment.
3. Start from the sample implementations in the starter notebook, improve them where you can, and document:
 - `detect_outliers_iqr(series, k=1.5)`
 - `detect_outliers_zscore(series, threshold=3.0)`
 - (Stretch) `winsorize_series(series, lower=0.05, upper=0.95)`
4. Apply to at least one numeric column; create a boolean outlier flag.
5. Sensitivity analysis (pick one model or summary):
 - Compare summary stats (mean/median/std) **with vs. without outliers**.
 - Fit a simple linear regression and compare coefficients/ R^2 /MAE **with vs. without outliers**; optionally include winsorized.
6. Reflection (≤ 1 page in Markdown cell):
 - Which method(s) and thresholds you chose and **why**.
 - Assumptions behind your choices.
 - Observed impacts on results.
 - Risks if assumptions are wrong.

Deliverables (due next class)

- A single Jupyter notebook with:

- Implemented functions and docstrings
- Sensitivity comparison (table and 1–2 plots)
- Reflection write-up

Grading Rubric (100 pts)

- (30) Correct, reusable functions with docstrings and parameterization
- (30) Sensitivity analysis comparing at least two treatments
- (20) Reflection on assumptions/risks tied to results
- (20) Code clarity, organization, and reproducibility (clean cells, seeded randomness)

Stretch Goals

- Implement winsorizing and re-run comparisons
- Visualize residuals and note any change in outlier influence
- Package functions into `src/outliers.py` and import them in your notebook

Example Expectations

- Clear boxplot before/after
- Table of metrics for three variants (all / filtered / winsorized)
- Reflection that ties method choice to the data's distribution and context